A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Principles of pollution control (CPA, CGM, new technologies) - Set 1

Mark scheme

AQA spec 3.4.3.1, 3.4.3.3; Pollution; textbook pp. 243-245, 303-304

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 2 points below. **1** Predicting the route / movement a pollutant will take through the environment (based on its properties and features of the environment) **1** To assess where it will become concentrated / which locations and organisms are most at risk, so monitoring can focus on those sites **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Sea current velocity and direction **1 Properties of the radionuclides:** half-life, solubility, tendency to adsorb onto sediments / bioaccumulate **1** Food chain pathways, eg uptake by seaweed, shellfish, fish eaten by people **1** Geology / hydrology of the coast – sediment deposition areas (mud flats) **1 Human activities in the area:** fishing, seaweed collection, beach use **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **half-life** → the time for the activity, or the amount, to fall by half • **adsorb** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3
01.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Identify the group of people whose lifestyle gives them the highest exposure (eg people who eat a lot of locally caught shellfish / fish, live near the mud flats, collect seaweed) **1** Monitor their exposure / the concentration of radioactive material in the foods they eat or their body **1** If their exposure / risk is acceptably low, everyone else is assumed to be at even lower risk – if not, emissions are controlled **1** CGM assesses exposure before health effects occur (not used to detect damage) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Critical Group** → the people most at risk / most heavily exposed	3

Do not write outside the box

Question	Marking guidance	Marks
01.4	**Any 2 of 7** 1 mark each for any 2 of the 7 points below, in any order, maximum 2. **1** Seawater **1** Sediment / mud **1** Seaweed **1** Shellfish / molluscs / fish **1** Milk, vegetables or meat from nearby farms **1** Atmospheric dust **1** Soil **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243, 303-304; AQA spec 3.4.3.1.	
02.1	**1 mark each** 1 mark for each of the 6 points below. **1 Location:** discharge into the sea / large water bodies where currents dilute and disperse the pollutant **1 Location:** chimneys / emissions sited downwind of urban areas so pollutants are not carried over populations – avoiding valleys and enclosed water bodies where pollutants concentrate **1 Location:** not discharging wastes onto permeable rock above an aquifer / not siting landfill on aquifers **1 Timing:** restrict polluting activities (industrial combustion, vehicle use) during temperature inversions when dispersal is poor **1 Timing:** discharge liquid effluent when river flow is high / on an ebb tide so it is diluted and carried away **1 Timing:** pesticide spraying at night / on still days – noisy operations restricted to daytime – night-flight restrictions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **permeable** → how easily water passes through / lets water through • **aquifer** → a permeable rock that stores groundwater / underground water store **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	6

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** The organisation / person that causes the pollution is responsible for (paying for) the damage caused **1** If the cost of preventing the pollution is less than the cost of the damage / fines, there is a financial incentive to prevent it – works best when penalties are high and escape is unlikely **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** It is assumed that the waste will cause pollution / harm until research shows it is unlikely to **1** So release is not permitted until it has been tested – lack of knowledge of a problem is not an excuse / does not reduce responsibility **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245; AQA spec 3.4.3.1.	
03.1	**1 mark each** 1 mark for each of the 2 points below. **1** Cost increases as the percentage prevented increases **1 The relationship is not linear:** cost rises slowly at first then very steeply / exponentially above about 80-90% **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.2	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 16.0 - 5.0 = 11.0 (£ million) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (£ million) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1

Do not write outside the box

Question	Marking guidance	Marks
03.3	**1 mark each** 1 mark for each of the 4 points below. **1 ALARA:** emissions should be As Low As Reasonably Achievable **1** It is rarely practical / affordable to reduce emissions to zero – low-level emissions may have acceptable impacts so an acceptable emission level is set **1 BATNEEC:** Best Available Technology Not Entailing Excessive Cost - choosing the most effective control equipment whose cost is not excessive **1** The factory would install equipment that gives a high level of control (eg 80-90%) rather than paying £16 million for 99% **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
03.4	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Industry / city may have expanded so there are more pollutant sources and total emissions have risen **1** Better understanding of the health / environmental effects so stricter standards are set **1** Improved technology becomes available (BATNEEC changes) / cost of better control falls **1** Pollutant may have accumulated (persistent) so background levels are higher **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistent** → does not break down / stays in the environment for a long time	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 244-245; AQA spec 3.4.3.1.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**1 mark each** 1 mark for each of the 4 points below. **1 Production prevention:** desulfurisation of fuels before combustion – low-temperature (fluidised bed) combustion to prevent NOx – smokeless fuels **1 Prevention of release:** electrostatic precipitators / bag filters / cyclones for smoke – catalytic converters – flue gas desulfurisation – bund walls – landfill liners **1 Post-release remediation:** oil spill clean-up (booms, skimmers, dispersants, bioremediation) – phytoremediation of metal-contaminated land – liming of acidified lakes **1 Alternative processes:** electric vehicles instead of petrol / diesel – renewable energy instead of fossil fuels – pyrethroids instead of organochlorines – fuel cells instead of internal combustion engines **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **desulfurisation** → removing sulfur dioxide from flue gases with lime or limestone **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	4
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Remediation is often incomplete / cannot recover all the pollutant (eg oil spills, dispersed pollutants), so some damage still occurs **1** Prevention avoids the damage entirely and is usually cheaper than clean-up / avoids polluter-pays costs – moving from post-production treatment to non-release is a circular economy principle **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
04.3	**1 mark each** 1 mark for each of the 4 points below. **1** Satellites detect and track the extent / movement of oil slicks over large areas, quickly and repeatedly, so clean-up can be directed and polluters identified **1** GPS tracking monitors ship positions so navigation errors / collisions / groundings are reduced and offshore shipping routes are enforced **1** Polymers with a high surface area adsorb metal ions from contaminated water / effluent so they can be removed and recovered **1** Plants that absorb heavy metals (hyperaccumulators) are grown on contaminated land, harvested and disposed of / metals recovered, cleaning the soil **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **adsorb** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 244, 303-304; AQA spec 3.4.3.1, 3.4.3.3.	
05.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** C **Also accept** • **Critical Group** → the people most at risk / most heavily exposed	1
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Monitoring can be focused on the sites predicted to be most at risk / where the pollutant will concentrate **1** No need to sample sites that were never likely to be polluted – sampling also checks the accuracy of the predictions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **accuracy** → close to the true value	2
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Behaviour of new chemicals can be predicted from similar existing chemicals: it is likely to persist, bioaccumulate and biomagnify **1** So acceptable emission limits, control methods and monitoring programmes can be set before release (precautionary principle) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistent** → does not break down / stays in the environment for a long time **Do not credit** • the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245; AQA spec 3.4.3.1.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** A waste is assumed to cause pollution until research shows it is unlikely to do so **Do not credit** • the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely	1
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** If the cost of preventing damage is less than the (penalty for) damage there is a financial incentive to prevent pollution **1** If polluters can escape their responsibilities / penalties are low, there is little incentive to comply **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2
06.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Marine discharges where currents dilute and disperse emissions **1** Emissions downwind of urban areas **1** Not discharging waste onto permeable rock above an aquifer **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **permeable** → how easily water passes through / lets water through • **aquifer** → a permeable rock that stores groundwater / underground water store **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-244; AQA spec 3.4.3.1.	
07.1	**1 mark each** 1 mark for each of the 2 points below. **1** Natural processes may concentrate the material (eg adsorption onto sediments, bioaccumulation and food-chain concentration) **1** It may be carried to locations where it is hazardous, such as fishing grounds or beaches used by people **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **adsorption** → nutrient ions sticking to the surfaces of clay and humus • **bioaccumulation** → a pollutant building up inside one organism over time **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2

Do not write outside the box

Question	Marking guidance	Marks
07.2	**Any 4 of 8** 1 mark each for any 4 of the 8 points below, in any order, maximum 4. **1** Physical state of the effluent (solution / solid) **1** Ocean currents / river flow **1** Effect of pH and oxygen on solubility **1** Bioaccumulation and food-chain concentration **1** Food sources and consumption by local people (eg shellfish, seaweed) **1** Half-lives of the isotopes **1** Meteorological conditions for any gaseous release **1** Nature of the environment - geology, vegetation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Bioaccumulation** → a pollutant building up inside one organism over time **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain	4
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Monitoring focuses on the sites most at risk of contamination **1** No need to sample sites that were never likely to be polluted – sampling checks the accuracy of the predictions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **accuracy** → close to the true value	2
07.4	**1 mark each** 1 mark for each of the 2 points below. **1** It assesses the risk to the people most exposed because of their lifestyle before any health effects occur **1** If their risk is acceptably low everyone else has a lower risk – if unacceptable, emissions are reduced to prevent future exposure **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Critical Group** → the people most at risk / most heavily exposed	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245, 303-304; AQA spec 3.4.3.1, 3.4.3.3.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 16.0 - 5.0 = £11 million **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (£ million) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
08.2	**1 mark each** 1 mark for each of the 3 points below. **1** It is rarely practical to reduce emissions to zero and the impact of low-level emissions may be acceptable, so an acceptable level must be decided **1 ALARA:** emissions should be As Low As Reasonably Achievable **1 BATNEEC:** the Best Available Technology Not Entailing Excessive Cost is chosen to achieve this **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** As industries expand or cities grow the number of pollutant sources increases **1** So the total emissions / combined concentration rises even if each source is unchanged, and higher standards are needed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
08.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Critical Pathway Analysis:** predicting movement of mobile persistent pollutants (state of effluent, wind, currents, geology, pH / oxygen and solubility, bioaccumulation, food sources, half-lives) so monitoring focuses on sites at risk – mainly radioactive discharges – checked by sampling • **Critical Group Monitoring:** exposure of the most-exposed group monitored before health effects – if their risk is acceptable everyone else is safer – emissions reduced if not • **Emission control strategies:** location (marine discharges, downwind of cities, not over aquifers) and timing (avoiding temperature inversions, low river flow) • **Principles:** polluter pays (incentive depends on penalties and enforcement), precautionary principle (assume harm until shown otherwise; ignorance no excuse), ALARA / BATNEEC and the non-linear cost-efficiency relationship • **Selection of methods:** production prevention (desulfurisation), prevention of release (precipitators, catalytic converters), post-release remediation (oil spill clean-up, phytoremediation), alternative processes (electric vehicles, renewables, pyrethroids) • **Evaluation:** prevention usually cheaper and more effective than remediation – costs rise steeply near 100% – CPA relies on understanding of pathways – enforcement and international agreement needed for transboundary pollutants – judgement about acceptable risk is political as well as scientific • Justified conclusion	9

Do not write outside the box

Question	Marking guidance	Marks
	Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **persistent** → does not break down / stays in the environment for a long time • **bioaccumulation** → a pollutant building up inside one organism over time • **Critical Group** → the people most at risk / most heavily exposed **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	
	Model answer - Level 3 (8 marks)	
	Deciding how much to control pollution starts with predicting where a pollutant will go. Critical Pathway Analysis uses the state of the effluent, wind and currents, geology, the effect of pH and oxygen on solubility, bioaccumulation and local food sources to predict whether a discharge will disperse harmlessly or concentrate somewhere sensitive; it is mainly used for radioactive discharges and means monitoring can focus on the sites at risk rather than sampling everywhere. Critical Group Monitoring then checks the exposure of the people most at risk from their lifestyle before any health effects occur: if their risk is acceptable everyone else's is lower, and if not, emissions are reduced.	
	The choice of control is guided by principles. The polluter pays principle gives a financial incentive to prevent damage, but only if penalties are high and polluters cannot escape responsibility. The precautionary principle assumes a waste is harmful until research shows otherwise, so ignorance is no defence. Because the cost of control rises steeply as emissions approach zero, emissions are set As Low As Reasonably Achievable using the Best Available Technology Not Entailing Excessive Cost. Where emission location and timing can be chosen (marine outfalls, downwind of towns, not over aquifers; avoiding temperature inversions) severity is reduced cheaply.	
	Overall, prevention of production (fuel desulfurisation) or of release (electrostatic precipitators, catalytic converters) is more effective than post-release remediation such as oil-spill clean-up, which never recovers all the pollutant. CPA depends on understanding pathways, and transboundary pollutants need international enforcement, but together these principles give a rational basis for deciding acceptable risk rather than aiming for an unaffordable zero.	
	Weaker answer - Level 1 (3 marks)	
	Pollution can be controlled in lots of ways. Critical Pathway Analysis looks at where pollution goes and Critical Group Monitoring looks at people. The polluter pays principle means the person who pollutes has to pay. The precautionary principle means being careful. Pollution can be prevented, or cleaned up afterwards with things like booms for oil spills. Electric cars are an alternative process. Pollution control can be expensive so ALARA is used. All of these are good ways of controlling pollution.	
	What makes the difference	

Do not write outside the box

Question	Marking guidance	Marks
	• The Level 3 answer explains *how* each principle works (why penalties matter, why costs rise near zero emissions); the weak answer only names them.	
	• The Level 3 answer groups the material logically (predict, decide, select) rather than listing everything at the same weight.	
	• 'Evaluate' needs comparison and a judgement: prevention vs remediation, CPA's dependence on understanding pathways, and a conclusion about acceptable risk. The weak answer ends with 'all of these are good'.	
	• Precise examples (desulfurisation, precipitators, catalytic converters, temperature inversions) show applied knowledge; 'things like booms' does not.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245, 303-304; AQA spec 3.4.3.1, 3.4.3.3.	

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